

Shyngyskhan Amangeldy

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Data Analyst and Systems Architect specializing in designing scalable backend infrastructure and big data analytics. Commercial experience with JSC Kazakhstan Temir Zholy acting as a Solutions Architect and Analyst. Academic R&D experience collaborating with Dr. Kamal Imran Mohd Sharif (PhD) on complex data analysis and co-authoring scientific papers. Co-founder of Tirek System and Solutions LLP.

Experience

Systems Architect & Lead Engineer | Tirek System and Solutions Ltd Dec 2024 – Present

- Designed and launched an advanced SaaS platform from scratch to accelerate academic research and big data analytics: **perricheno.ru**, strategically planned for scaling via Astana Hub Startups. Core concept: Research, write, analyze, cite, and visualize inside a unified ecosystem.
- DevOps & Architecture:** Built a fault-tolerant microservice architecture on Linux. Containerized all internal services using Docker and established a robust CI/CD pipeline (GitHub Actions) featuring **Zero-Downtime Deployment**, Cloudflare integration, and a custom Uptime monitoring system.
- LLM Agents:** Developed an asynchronous, multi-layered agentic system utilizing a 6-stage sequential generation flow. Resource-intensive document drafting executes server-side autonomously via isolated user session management and background task processing.
- Data Analytics & Compilers:** Integrated securely isolated compiler environments. The platform ingests datasets comprising millions of rows, allowing AI agents to write and execute **R / Python** scripts for advanced statistical analysis and generate highly precise, editable visualizations across 40+ distinct chart types.
- RAG & NLP:** Implemented semantic search engines integrated directly with arXiv and OpenAlex APIs, featuring automated generation of BibTeX citations and technical scientific diagrams (TikZ / LaTeX).

ML / Solutions Architect | JSC Kazakhstan Temir Zholy (Pre-IPO / Public Enterprise) Feb 2024 – May 2024

- Logistics Optimization Service:** Co-developed an enterprise-grade platform to reduce 8–12 hour locomotive idle times across a rail network spanning 1,000+ stations for a pre-IPO joint-stock enterprise. Implemented a production **ML model** to automatically match and allocate optimal candidate resources to 90 distinct locomotives.
- Spatial Data & Simulation:** Integrated interactive web mapping layers leveraging OpenStreetMap (OSM) and OpenRailwayMap. Designed an isolated sandbox environment to simulate dispatch schedules, operational edge cases, and emergency scenarios. Projected savings estimated at **\$80–\$100 / hour** (scaling up to approx. **\$1,000** saved per locomotive).
- Incident Automation:** Architected an automated ticketing and complaint handling workflow featuring two strictly segregated permission rings for external clients and corporate staff. Configured automated request routing to enterprise line managers via custom Telegram bots and Email distribution pipelines.

Projects & Academic Research

E-Commerce Logistics Research | Co-Author w/ Dr. Kamal Imran Mohd Sharif Jan 2026 – Feb 2026

- Co-authored a comprehensive quantitative research project on Last-Mile fulfillment analytics under the mentorship of Professor Dr. Kamal Imran Mohd Sharif (PhD in Technology, Operation and Logistics, Universiti Utara Malaysia).
- Processed and analyzed a massive transactional database exceeding **10,000,000+** records in **R**, applying rigorous **Binary Logistic Regression** models to evaluate the operational trade-off between absolute delivery speed and SLA reliability metrics.

Technical Skills

- Programming Languages:** Python, R.
- Databases & Analytics:** PostgreSQL, Pandas, Matplotlib, Seaborn, Power BI, Tableau.
- Backend & Architecture:** Microservices, Multi-Agent Systems, RAG Architectures, NLP (spaCy).
- DevOps & Cloud:** Linux (Ubuntu/Debian server administration), Google Cloud Platform (Compute Engine, VPC, VDS), Docker, CI/CD Automation (GitHub Actions), Cloudflare.
- Tools & Geospatial:** Git / GitHub, OpenStreetMap (OSM), LaTeX, TikZ, Figma.

Education

Astana IT University | Bachelor's Degree in AI & Business 2024 – 2027 (Expected)

Languages

Russian (C1), Kazakh (Native), English (Upper-Intermediate / B2 Technical)